Queen Mary
University of London
Science and Engineering

EBU4202: Digital Circuit Design
Karnaugh Maps Tutorials (Part 1)

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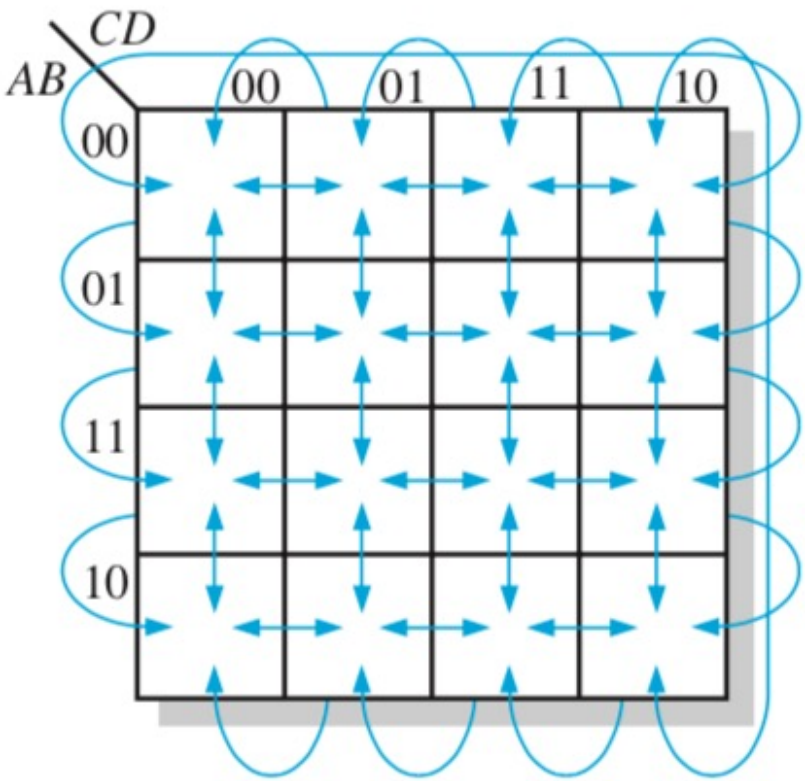
Cell Adjacency



Wraparound

Additional: K-Map with Prime Implicants

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	1	1
10	1	0	1	1

Additional: K-Map with Prime Implicants

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	1	1	1
10	1	0	1	1

Good approach: Use the fewest possible number of maximal groupings needed to cover all of the squares marked with a 1.

$$F(A, B, C, D) = BD + CD + AC + B'D'$$

Additional: Essential Prime Implicants

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	1	1	1
11	1	0	1	1
10	0	0	1	0

Additional: Essential Prime Implicants

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	1	1	1
11	1	0	1	1
10	0	0	1	0

F(A, B, C, D) = A'D + CD + BD'

Exercise 1: Logic Function Minimization

Simplify, using a Karnaugh map:

F(X, Y, Z) = X’ + XY’ + XYZ’

X \ YZ	00	01	11	10
0				
1				

Exercise 1: Logic Function Minimization

Step 1: Build the Karnaugh map:

F(X, Y, Z) = X’ + XY’ + XYZ’

000	100	110
001	101	
010		
011		

X \ YZ	00	01	11	10
0	1	1	1	1
1	1	1	0	1

Continue…..

Exercise 2: Logic Function Minimization

Determine the minimum SOP from the Karnaugh map:

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	1	1
11	1	1	1	1
10	0	1	0	0

Exercise 3: Logic Function Minimization

Determine the minimum SOP from the Karnaugh map:

AB \ CD	00	01	11	10
00	0	1	0	0
01	0	1	1	0
11	1	1	1	1
10	1	1	0	0

Exercise 4: Logic Function Minimization

Determine the minimum SOP from the Karnaugh map:

AB \ CD	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	1	0
10	1	0	1	0

Exercise 5: Logic Function Minimization

Determine the minimum SOP from the Karnaugh map:

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	0	1
11	0	1	1	1
10	0	0	0	0

F= A'BC' + ABD + BCD'

PI

AB \ CD	00	01	11	10
00	0	0	0	0
01	1	1	0	1
11	0	1	1	1
10	0	0	0	0

F= BC'D + ABC + A'BD'

PI

NO EPI

Relationships: F, F′, Σ, Π

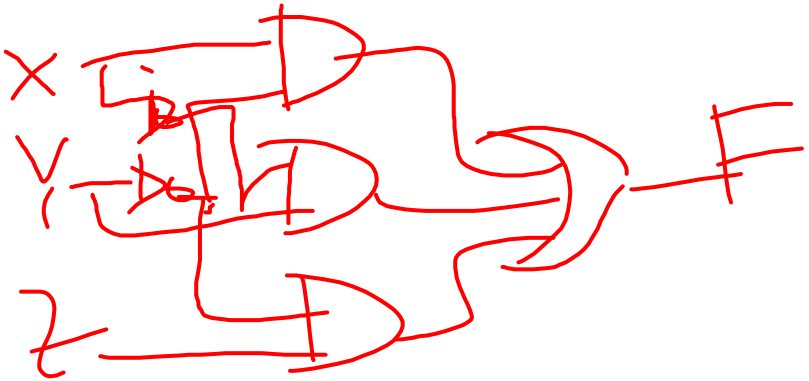
Find the SoP and PoS for the following K-Map.

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	0	1
11	1	0	0	0
10	1	1	0	0

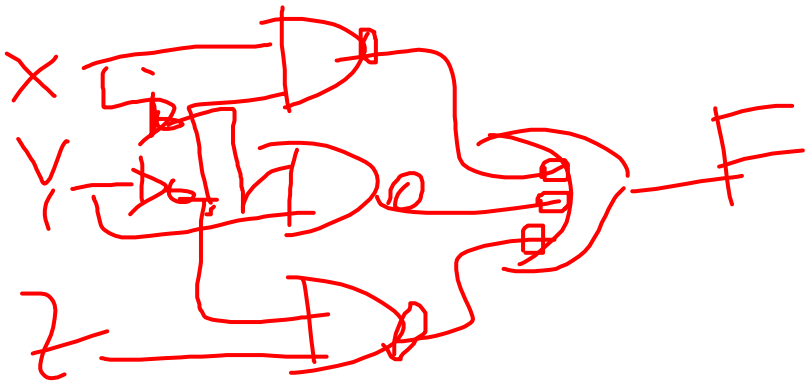
Class Example: SOP of F *To be completed in class …*

- Draw the *NAND gate* implementation of: F = XY′ + X′Y + Y′Z

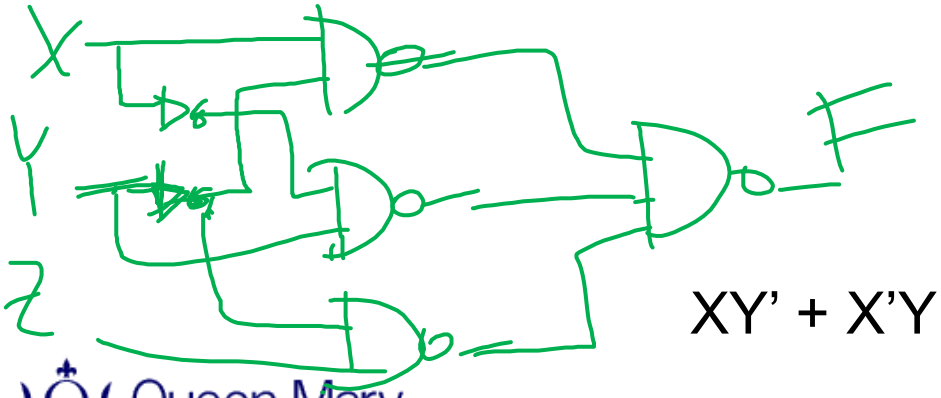
✓ 1. Draw the normal circuit:



✓ 2. Apply De Morgan's theorems graphically:



✓ 3. Convert the 3-inverted input gate to *NAND*:



XY' + X'Y + Y'Z = ((XY')')' + ((X'Y)')' + ((Y'Z)')'
= [(XY')'.(X'Y)'.(Y'Z)']'

NAND gate as NOT gate

Strictly speaking, ALL NAND implementation should not include a NOT gate.

So a NAND gate should be used as an inverter.

Looking at the truth table of a NAND gate:

A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

We see that if the inputs are connected together then the output is the inverse of the input.
So a NAND gate with the inputs connected together can be use as a NOT gate.

Example: Don't-Care Cases - PoS

- Design a circuit that will produce an output 1 if the BCD input combination is an even number. The BCD input combination 0000 is not allowed to occur.

(1) Find the PoS using the Karnaugh map.

(2) Find the circuit diagram

#	A	B	C	D	F(A,B,C,D)
0	**0**	0	0	0	
1	**0**	0	0	1	
2	**0**	0	1	0	
3	**0**	0	1	1	
4	**0**	1	0	0	
5	**0**	1	0	1	
6	**0**	1	1	0	
7	**0**	1	1	1	
8	**1**	0	0	0	
9	**1**	0	0	1	
10	**1**	0	1	0	
11	**1**	0	1	1	
12	**1**	1	0	0	
13	**1**	1	0	1	
14	**1**	1	1	0	
15	**1**	1	1	1	

Example: Don't-Care Cases - SoP

Simplify:

$F(A, B, C, D) = \Sigma m(1, 3, 5, 7, 9) + d(6, 12, 13)$

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	x
11	x	x	0	0
10	0	1	0	0

Class Example:
3-Bit Even Parity Generator

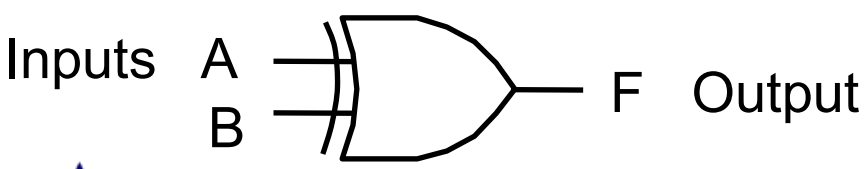
To be completed in class …

Truth Table

X	Y	Z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

X \ YZ	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$\mathbf{F} = XY'Z' + XYZ + X'Y'Z + X'YZ'$$
$$= X(Y'Z' + YZ) + X'(Y'Z + YZ')$$
$$= X(Y \oplus Z)' + X'(Y \oplus Z)$$
$$= \mathbf{X \oplus Y \oplus Z}$$



A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

$F = (A \oplus B)$ *or*

$F = A'B + AB'$ *or*

$F = \bar{A}B + A\bar{B}$

$F' = AB + A'B'$

5-Variable Karnaugh Map

Exercise: ***Using a 5-variable Karnaugh Map, find the minimal PoS for the Boolean function:***

$$F(A, B, C, D, E) = \Sigma m(1, 3, 5, 8, 10, 12, 16, 20, 21, 25) + d(0, 4, 7, 28, 30)$$

A = 0

BC \ DE	00	01	11	10
00	0: x	1: 1	3: 1	2: 0
01	4: x	5: 1	7: x	6: 0
11	12: 1	13: 0	15: 0	14: 0
10	8: 1	9: 0	11: 0	10: 1

A = 1

BC \ DE	00	01	11	10
00	16: 1	17: 0	19: 0	18: 0
01	20: 1	21: 1	23: 0	22: 0
11	28: x	29: 0	31: 0	30: x
10	24: 0	25: 1	27: 0	26: 0